

Assignment 2

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Task 1

To complete this task, first I loaded the tidyverse library.

```
library(tidyverse) #load tidyverse library
```

Then, I used the readr package to load the college.data.csv file into an object called college_data with the read_csv() command

```
college_data <- read_csv("./data/collegedata.csv") #load collegedata.csv
```

```
Rows: 7175 Columns: 12
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (4): MN_EARN_WNE_P6, INSTNM, GRAD_DEBT_MDN, AGE_ENTRY
```

```
dbl (8): UNITID, OPEID, SAT_AVG, ADM_RATE, UGDS, COSTT4_A, AVGFACSAL, ICLEVEL
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Task 2

Even though the read_csv() command gives you a message with the number of rows and columns, you can also use the dplyr command glimpse() to see these details:

```
glimpse(college_data) #view the rows and column information for the dataframe
```

```

Rows: 7,175
Columns: 12
$ UNITID      <dbl> 100654, 100663, 100690, 100706, 100724, 100751, 100760,~
$ OPEID       <dbl> 100200, 105200, 2503400, 105500, 100500, 105100, 100700~
$ MN_EARN_WNE_P6 <chr> "27800", "37600", "39400", "41300", "23500", "38500", "~
$ INSTNM      <chr> "Alabama A & M University", "University of Alabama at B~
$ SAT_AVG     <dbl> 849, 1125, NA, 1257, 825, 1202, NA, NA, 1009, 1217, 115~
$ ADM_RATE    <dbl> 0.8738, 0.5814, NA, 0.7628, 0.4590, 0.5259, NA, NA, 0.7~
$ UGDS        <dbl> 4616, 12047, 293, 6346, 4704, 31663, 1492, 2888, 4171, ~
$ COSTT4_A    <dbl> 22667, 22684, 13380, 22059, 19242, 28422, 13868, NA, 19~
$ AVGFACSAL   <dbl> 7028, 10517, 3857, 9463, 7952, 9802, 5960, 8367, 7251, ~
$ GRAD_DEBT_MDN <chr> "32750", "21833", "22890", "22647", "31500", "23290", "~
$ AGE_ENTRY   <chr> "20.28374137", "23.60797466", "33.6722973", "22.7279196~
$ ICLEVEL     <dbl> 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 2, 1, 1, 2, 2, 1, 2, 3~

```

Using either method, we find out that there are 7175 rows and 12 columns in this dataframe.

Task 3

To create a new dataframe with only certain columns, we can use the `dplyr` command `select()` to choose the requested columns: `INSTNM`, `SAT_AVG`, and `ADM_RATE`.

```

#create a new dataframe that has the requested columns
admission_rates <- college_data |> select(INSTNM, SAT_AVG, ADM_RATE)

```

I like to confirm that everything worked as I wanted by running the `head()` or `glimpse()` command to verify.

```

#confirming the new dataframe:
glimpse(admission_rates)

```

```

Rows: 7,175
Columns: 3
$ INSTNM      <chr> "Alabama A & M University", "University of Alabama at Birming~
$ SAT_AVG     <dbl> 849, 1125, NA, 1257, 825, 1202, NA, NA, 1009, 1217, 1150, NA,~
$ ADM_RATE    <dbl> 0.8738, 0.5814, NA, 0.7628, 0.4590, 0.5259, NA, NA, 0.7659, 0~

```

Task 4

First, I created a new dataframe with the NA rows removed, using the `drop_na()` command from the `tidyr` package, and `glimpse()` to confirm my changes.

```
cleaned_data <- admission_rates |> drop_na() #drop rows with NA values
glimpse(cleaned_data) #confirm dropped values
```

Rows: 1,301

Columns: 3

\$ INSTNM <chr> "Alabama A & M University", "University of Alabama at Birming~

\$ SAT_AVG <dbl> 849, 1125, 1257, 825, 1202, 1009, 1217, 1150, 972, 1001, 1062~

\$ ADM_RATE <dbl> 0.8738, 0.5814, 0.7628, 0.4590, 0.5259, 0.7659, 0.8054, 0.483~

Then, I used `ggplot()` from the `ggplot2` library to create a scatter plot of Average SAT scores versus Admission Rates. I styled it a bit with blue points and the black and white theme, and added axes labels and a title and subtitle.

Interestingly, it took me a minute to figure out what this data was telling me, but when I thought about it, it makes sense. The higher the school's average SAT score, the lower their admission rate due to their high selectivity.

```
#create a new plot of SAT scores versus admission rate
ggplot(cleaned_data, aes(x = SAT_AVG, y = ADM_RATE)) + #create the plot
  geom_point(color="blue") +
  labs(x = "Average SAT Score", #apply labels
       y = "Admission Rate",
       title = "Average SAT Scores versus Admission Rates",
       subtitle = "Each point is an individual college or university") +
  theme_bw() #apply theme
```

Average SAT Scores versus Admission Rates

Each point is an individual college or university

